STDGX FINAL REPORT: BORNEO HACK WEEKEND

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1.0 Introduction

1.1 Background of Case Study

Lovenia et al. (2024) explained that Southeast Asia (SEA) is one of the most linguistically diverse regions in the world. More than 1,300 local languages are spoken by around 671 million people. However, even with this rich language diversity, many modern AI systems do not work well for SEA languages because there is not enough data available in text, images, and speech. Most AI systems are mainly trained using English data, which makes it difficult for them to understand SEA languages properly. This can also lead to cultural misunderstandings and bias in AI systems.

While this issue affects the broader Southeast Asian region, this project focuses specifically on Malaysia. Malaysian indigenous languages are divided into two language families. The first is the Austronesian family, which is the largest group in terms of both the number of speakers and the number of languages that can be found across Malaysia. The second one is Austroasiatic family, consisting of about 20 languages, these languages are uniquely used by the aborigines, or Orang Asli (Omar, 2014).

A clear example of this digital gap can be seen in Sarawak, where local dialects like Sarawak Malay are widely spoken and are an important part of the region's culture. Despite its everyday use, Sarawak Malay is rarely included in digital tools, language learning apps, or AI systems. This limits opportunities for people outside Sarawak, such as tourists or learners from other regions, to easily learn and use the language, which could help them communicate while visiting and gain better understanding of local culture. At the same time, providing digital support can also benefit local communities, especially younger generations, by helping them improve reading and writing skills and keeping the language visible in modern digital spaces. By targeting both outsiders and locals, digital solutions can both promote cultural exchange and support language preservation.

1.2 Chosen Sustainable Development Goal (SDG)

This project aligns primarily with United Nations Sustainable Development Goal 4: Quality Education, specifically:

1. 4.6 – Ensure that all youth and a substantial proportion of adults achieve literacy and numeracy.
2. 4.7 – Promote education for sustainable development, including cultural awareness and global citizenship. By developing a technology-based language preservation tool for Sarawak Malay, the project supports inclusive and equitable quality education while promoting cultural and linguistic diversity.

The project also supports SDG 10: Reduced Inequalities, specifically:

3. 10.2 – Empower and promote the social, economic, and political inclusion of all individuals, regardless of linguistic or cultural background. It helps reduce the digital gap faced by speakers of underrepresented dialects.

Additionally, the project connects to SDG 11: Sustainable Cities and Communities, specifically:

4. 11.4 – Strengthen efforts to protect and safeguard cultural heritage. It contributes to preserving Sarawak Malay as part of Malaysia's cultural heritage in the digital era.

1.3 Problem Statement

Sarawak Malay is a dialect of Malay mainly spoken in Sarawak. It has its own vocabulary, pronunciation, and expressions that are quite different from Standard Malaysian Malay. The dialect also varies across different regions of Sarawak, influenced by local languages such as Iban, Bidayuh, Kedayan, and Brunei Malay (Asmah Hj Omar, 2008, cited in Sarbini-Zin et al., 2019). Although it is commonly used in daily conversations and within local communities, it is rarely documented in formal language resources or included in digital platforms. Currently, there are very few digital tools that support Sarawak Malay. There are no proper translation apps, online dictionaries, or AI chat systems designed specifically for this dialect. Because of this, speakers of Sarawak Malay do not have the same digital access as users of more widely supported languages.

The lack of digital support for Sarawak Malay creates several important problems. For people outside Sarawak, such as tourists or learners from other regions, it is difficult to learn and use the language, limiting their ability to communicate and engage with the local culture. For local communities, especially younger generations who rely heavily on technology, the absence of apps, AI tools, and other digital platforms may cause the dialect to lose visibility and everyday use. Without proper digital representation, Sarawak Malay risks declining in both educational and online spaces, which could threaten its cultural value and long term survival.

1.4 Objectives

1. Develop an AI-powered tool that can translate Standard Malay and English into Sarawak Malay, helping people use the dialect digitally.
2. Gather and organize words, phrases, and common expressions in Sarawak Malay, making it available for anyone to use in learning, research, or AI development.
3. Create user friendly platform where both local and general can learn, practice, and use Sarawak Malay in an interactive way.
4. Increase the presence of Sarawak Malay in digital and educational spaces, helping to preserve the language and culture while supporting the United Nations Sustainable Development Goals (SDG 4, SDG 10 & SDG 11).

2.0 Review of Existing Solutions

2.1 Overview

In recent years, people have been paying more attention to ways of preserving minority and low-resource languages using digital tools. Around the world, different methods have been tried, such as AI translation tools and language learning apps that make learning more accessible. However, there are very few solutions that support Sarawak Malay, showing a clear gap and the need for this project.

2.2 AI Translation Tools

Large-scale multilingual language technologies such as Meta's No Language Left Behind (NLLB) model have contributed to expanding digital language coverage with the goal of eradicating language barriers on a global scale. Released in 2022, the NLLB model supports over 200 languages and demonstrates improvements in translating low resource languages through multilingual training approaches (Team et al., 2022). Even with these improvements, local dialects and everyday versions of language, like Sarawak Malay, are still mostly left out of these systems. The difference between a standard language and a spoken dialect is still a problem, because most language technologies focus on official languages instead of the different ways people actually speak.

2.3 Mobile Language Learning Application

Language learning apps have become a popular way for people to use technology to improve their vocabulary and language skills. Two of the most well-known apps are Duolingo and Babbel, and they are popular because they use very different teaching methods that can help almost any type of learner (Riaz & Nayab, 2025). Duolingo offers about 95 languages to choose from and allows users to learn in languages other than English (Shortt et al., 2021), while Babbel provides 13 languages for learning according to its official website. However, these apps do not support Sarawak Malay or other Malaysian regional languages. These apps rely on large amounts of existing language data and a big group of users to make the development costs worthwhile. Because of this, smaller language communities, like those who speak Sarawak Malay, are often left out from modern tools for learning and communication.

2.4 Gaps identified in Existing Solutions

1. Lack of support for Sarawak Malay
    - AI translation tools like NLLB and popular language learning apps like Duolingo and Babbel do not include Sarawak Malay or other Malaysian regional dialects.
    - This shows that regional and minority languages are often overlooked even when larger systems aim to cover many languages.
2. Focus on standard or official languages
    - Current AI models and apps prioritize formally recognized languages rather than the variations within a language, like dialects or everyday spoken forms. This leaves a gap for people who speak local versions of a language.
3. Limited access and usability for smaller language communities
    - Apps require large datasets and many users to justify development. Small language communities often do not have this, so they are excluded from AI learning tools.
    - This gap affects not just learning but also preservation and wider accessibility of the language.
4. Gap between spoken and written forms
    - Even if people already speak Sarawak Malay, most tools don't help them read, write, or document it, which makes it harder to preserve the language.

2.5 Summary

The review of existing solutions shows that, while AI translation tools like NLLB and popular language learning apps such as Duolingo and Babbel have made progress in supporting many languages, none of them include Sarawak Malay and other regional Malaysian dialects. Current tools focus mainly on official or widely spoken languages and often fail to handle the differences between standard languages and local spoken dialects.

Smaller language communities face additional challenges because most apps and AI models require large datasets and many users to justify development, leaving regional languages underrepresented. Furthermore, even when people already speak a dialect, there is little support for reading, writing, or documenting it, which affects both language learning and preservation. These gaps highlight a clear need for solutions that specifically support Sarawak Malay, making it more accessible for both local users and outsiders, and helping to preserve

the language for future generations. Currently, the solution focuses on Sarawak Malay, but it is designed to evolve step by step to include other indigenous languages in the future, increasing its accessibility and relevance across multiple ASEAN countries.

3.0 Methodology (Non-Technical)

This project focuses on designing a mobile application prototype called SENADA that helps users learn Sarawak Malay while also introducing the cultural context behind the language. The goal of the prototype is to demonstrate how digital technology can support regional language learning while preserving local cultural knowledge.

The prototype was developed using Figma, which allows designers to create realistic mobile interfaces and simulate navigation between screens. This approach helps visualize the application flow before the actual system development.

During the design process, the team applied a design thinking approach which involves understanding the problem, identifying user needs, designing possible solutions, and evaluating the usability of the prototype. This approach ensures that the application is designed from the user's perspective and addresses real challenges faced by learners who want to understand Sarawak Malay.

The proposed solution focuses on supporting Sarawak Malay language learning while preserving local culture. The prototype integrates learning features, cultural stories, and interactive activities that help users engage with the language in a practical and meaningful way.

In addition, the design ensures good user experience (UX) by maintaining consistent colors, typography, and layout throughout the application. The interface also uses clear language and readable contrast to ensure accessibility for different users.

The application also considers ASEAN inclusivity, as it supports language learning for regional dialects that are often underrepresented in digital platforms. By focusing on Sarawak Malay, the prototype highlights the importance of preserving regional languages while making them accessible to a wider audience.

The methodology used in this project was designed to support the objectives of the SENADA platform. The prototype development demonstrates how an AI-powered translation tool can help translate Standard Malay and English into Sarawak Malay, allowing users to interact with the dialect in digital environments. The feature design also organizes Sarawak Malay vocabulary, phrases, and expressions into structured learning modules and cultural stories,

making the language more accessible for learning and reference. In addition, the platform was designed to be user-friendly and interactive through lessons, quizzes, and storytelling features so that both local users and general learners can practice the language easily. By integrating language learning with digital technology and cultural content, the project also aims to increase the presence of Sarawak Malay in digital and educational spaces while supporting the United Nations Sustainable Development Goals related to quality education, reduced inequalities, and sustainable communities.

3.1 System Design Approach

The system was designed to make language learning simple, engaging, and culturally meaningful. The application is divided into several main sections such as Homepage, Learn, Translate, Explore Stories, Game & Quiz, Community, Setting and Profile.

Each section serves a specific purpose to support the learning process. The Learn page focuses on teaching vocabulary and common phrases, while the Translate page helps users understand unfamiliar words. The Explore Stories section introduces cultural stories from Sarawak, allowing users to learn about traditions and local heritage.

Interactive elements such as games and quizzes were included to make the learning experience more enjoyable and motivating. These features allow users to practice their knowledge while earning rewards.

The interface design also incorporates cultural elements inspired by Sarawak, such as traditional patterns and colors, to create a unique identity for the application.

This modular structure allows users to move between learning, translation, cultural exploration, and interactive activities smoothly, creating a complete language learning ecosystem within the application.

3.2 Prototype Development

The prototype was developed using Figma, a design platform that allows the team to create interactive mobile application screens.

Several pages were designed, including the Homepage, Learn Page, Translate Page, Explore Story Page, Game & Quiz Page, Coin Reward System, and Subscription Page. These pages are connected through interactive links to simulate how users would navigate through the application.

The prototype demonstrates a working user flow, allowing users to move between pages such as selecting lessons, translating words, reading cultural stories, and playing quizzes.

This approach helps demonstrate the functionality and usability of the application without needing to build the full system. It also allows the team to evaluate the design logic, user experience, and feature integration.

3.3 Feature Design

The application includes several features designed to support both language learning and cultural awareness.

The Learn feature provides structured lessons such as greetings, daily conversations, common phrases, and time expressions in Sarawak Malay. These lessons help users gradually build their vocabulary and communication skills.

The Translate feature allows users to translate words between Sarawak Malay, Standard Malay, and English. This helps users understand unfamiliar words and improves communication with local speakers.

The Explore Stories section introduces cultural knowledge through categories such as Iban Culture, Folk Tales, Legends, and Festivals. These stories help users understand Sarawak traditions while learning new vocabulary.

The application also includes Game & Quiz activities, where users can test their knowledge through interactive questions. Completing quizzes allows users to earn coins, which can later be converted into vouchers through the reward system.

This combination of learning, storytelling, and gamification makes the application more engaging and encourages users to continue learning.

The SENADA application also introduces several unique elements that differentiate it from existing language learning applications. Unlike common language learning platforms that focus mainly on widely spoken languages, SENADA specifically focuses on Sarawak Malay, which is rarely supported in digital learning platforms. In addition, the application integrates AI-assisted podcast storytelling and a coin-to-voucher reward system to encourage continuous learning. These features provide a more interactive and culturally meaningful learning experience.

In addition, the application integrates AI-assisted features such as podcast storytelling and enhanced translation explanations. These features provide users with richer learning content by combining audio learning, contextual explanations, and cultural storytelling. This integration of AI technology improves the learning experience by making language learning more interactive and engaging.

3.4 User Access Levels (Guest, Basic, Premium)

The application was designed with three different user access levels to provide flexibility and encourage users to explore the platform.

Guest users can explore the application with limited features. They can view some story categories and use basic translation features, but they are required to sign up to access full content or save vocabulary.

Basic Plan users have access to most of the learning features such as lessons, translation, quizzes, vocabulary saving and also coin-to-voucher conversion. They can read full cultural stories but do not have access to advanced features such as podcasts or camera translation.

Premium Plan users have full access to all features. This includes camera translation, podcast storytelling, the ability to add words or phrases to favourites, and coin-to-voucher conversion. Premium users can also earn additional rewards through learning activities.

This tiered access system allows users to experience the application first before deciding to upgrade to premium features.

3.5 AI Integration

Artificial Intelligence (AI) is integrated into the SENADA application to enhance the language learning experience. AI technology supports several key features including translation assistance, pronunciation guidance, and podcast storytelling.

For the translation feature, AI can assist in generating translation suggestions between Sarawak Malay, Standard Malay, and English. When a user enters a word or sentence, the AI system processes the input and provides the translated output together with explanations that help users understand the meaning more clearly.

The system then generates translation suggestions, pronunciation guidance, or podcast narration depending on the feature being used.

AI can also support pronunciation guidance by helping learners understand how certain Sarawak Malay words are pronounced. This is particularly useful for beginners who are not familiar with the dialect.

The use of AI in this application is appropriate because it helps automate language processing tasks such as translation suggestions and audio generation while improving user interaction with the learning content.

The design of the SENADA application also considers ethical use of AI and user data privacy. The system does not collect sensitive personal information from users. AI processing only uses the text input provided by users for translation or learning purposes. This ensures that user data remains protected while using the application.

The system design also considers potential bias in AI-generated outputs, and the results are reviewed to ensure that translations and explanations remain accurate and culturally appropriate.

4.0 Results

This section summarizes the outcome of the prototype development for the SENADA mobile application. The prototype demonstrates how the proposed solution can support Sarawak Malay language learning while also introducing cultural knowledge from the region.

The prototype includes several key features such as the Homepage, Learn Page, Translate Page, Explore Story Page, Game & Quiz Page, and Coin Reward System. These features were designed to create an interactive and engaging learning environment for users.

Through the prototype, users are able to navigate between different sections of the application and experience the overall learning flow. This includes learning vocabulary through structured lessons, translating words between languages, reading cultural stories, participating in quizzes, and earning rewards through coins.

Overall, the prototype successfully demonstrates how the application can function as a digital learning platform for Sarawak Malay and shows how the proposed solution meets the project objectives of improving language accessibility and promoting cultural awareness.

4.1 Problem-Solution Fit

The prototype directly addresses the main problem identified in the case study, which is the lack of digital tools that support Sarawak Malay and its cultural knowledge.

Currently, many language learning applications focus only on widely spoken languages and do not include regional dialects such as Sarawak Malay. As a result, learners and visitors who want to understand the dialect have very limited resources.

The SENADA prototype provides a solution by offering a platform that integrates language lessons, translation tools, and cultural stories in one application. This approach helps users learn Sarawak Malay while also understanding the cultural background of the language.

By making the language available in a digital format, the application contributes to preserving Sarawak Malay and increasing its visibility in modern digital spaces.

4.2 Creative Approach

One of the unique aspects of the SENADA application is the combination of language learning with cultural storytelling. Instead of focusing only on vocabulary lessons, the application introduces stories related to Sarawak traditions and heritage.

The Explore Story feature includes categories such as Iban Culture, Folk Tales, Legends, and Festivals. These stories allow users to learn cultural knowledge while also improving their vocabulary.

In addition, the application includes interactive elements such as quizzes and games. These features make the learning process more engaging and enjoyable for users.

The reward system also adds a gamification element to the application. Users can earn coins by completing quizzes and learning activities, and these coins can later be converted into vouchers. This encourages users to stay motivated and continue learning.

4.3 ASEAN Relevance

Although the prototype focuses specifically on Sarawak Malay, the concept can be expanded to support other indigenous languages across the ASEAN region.

Many Southeast Asian languages face similar challenges due to limited digital resources and lack of representation in technology platforms. By creating a platform that supports regional languages, the SENADA application can help promote linguistic diversity in the region.

In the future, the system could potentially include other local languages from ASEAN countries, making the platform more inclusive and accessible for a wider audience.

In many ASEAN countries, local languages face similar challenges such as limited digital documentation and lack of technological support. The SENADA prototype demonstrates how digital learning platforms can help address these constraints while promoting linguistic diversity across the region.

This approach demonstrates how digital learning platforms can support the preservation of indigenous languages across ASEAN while improving access to regional cultural knowledge.

4.4 Commercial Viability

The SENADA application also demonstrates potential for commercial use through its subscription model.

The platform includes three types of user access levels which are Guest Users, Basic Plan Users, and Premium Plan Users. Guest users can explore the application with limited features, while Basic Plan users have access to most learning functions including coin-to-voucher rewards.

Premium Plan users can unlock advanced features such as podcast storytelling, voice and camera translation, and the ability to save unlimited words or phrases in the favourites section. Premium users also receive additional rewards such as bonus game coins, extra daily challenge coins, and a premium badge.

This subscription structure allows the application to generate revenue while still providing free learning opportunities for new users.

Additionally, the reward system encourages continuous user engagement, which can help increase user retention and long-term usage of the platform.

This freemium approach allows the platform to attract new users while generating revenue from advanced features offered through the premium subscription.

4.5 Prototype Interface Screens

This section presents several key interface screens from the SENADA mobile application prototype. These screens demonstrate how users interact with the main features of the application. The prototype was designed to provide a simple and user-friendly interface that allows users to easily navigate between different learning features.

4.5.1 Learn Page

Figure 1: Learn Page

The Learn Page provides structured lessons that help users understand basic Sarawak Malay vocabulary and phrases. The lessons are organized into several topics such as greetings, daily conversations, common phrases, and time expressions.

Each lesson includes commonly used words or phrases together with their meanings, allowing users to gradually build their vocabulary and communication skills. Users can also listen to the pronunciation of each phrase through the audio feature and save useful words to their favourites for future reference.

The layout is designed to be simple and easy to navigate so that users can focus on the learning content and progress through the lessons step by step.

4.5.2 Translate Page

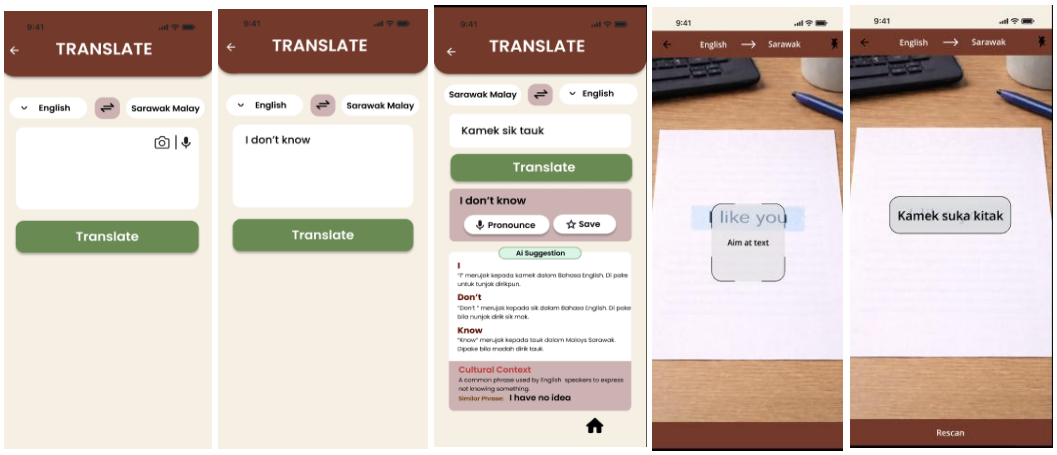

Figure 2: Translate Page

The Translate Page allows users to translate words and simple sentences between Sarawak Malay and English. This feature helps users understand unfamiliar vocabulary and supports communication between speakers of different languages.

Users can enter text directly into the translation field or use the camera translation feature to scan and translate text from images. The system then provides the translated result together with pronunciation guidance and additional explanations that help users better understand the meaning and usage of the phrase.

Basic Plan users can perform standard text translations, while Premium Plan users have access to advanced features such as camera translation and more detailed AI-assisted explanations that provide contextual and cultural information related to the translated phrase.

4.5.3 Explore Story Page

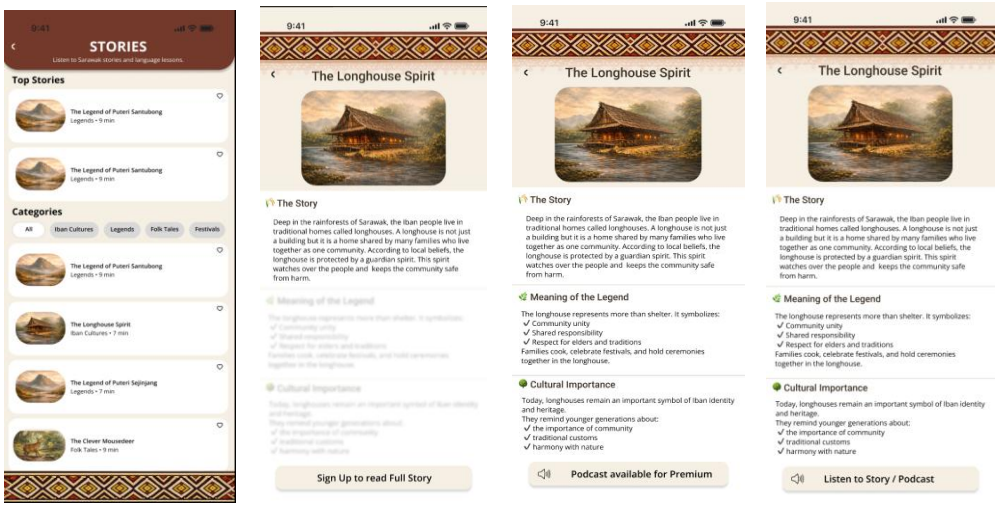

Figure 3: Explore Story Page

The Explore Story Page introduces cultural stories from Sarawak that help users learn the language while understanding local traditions and heritage. The stories are organized into several categories such as Iban Culture, Folk Tales, Legends, and Festivals, allowing users to easily browse different types of cultural content.

Each story includes a description of the story, the meaning behind the legend, and the cultural importance related to Sarawak traditions. This feature allows users to gain cultural knowledge while improving their Sarawak Malay vocabulary.

Guest users can explore the story categories but must sign up to read the full story content. Basic Plan users can access and read the full stories available in the application, while Premium Plan users receive additional features such as podcast narration that allows them to listen to the stories in audio format.

4.5.4 Game & Quiz Page

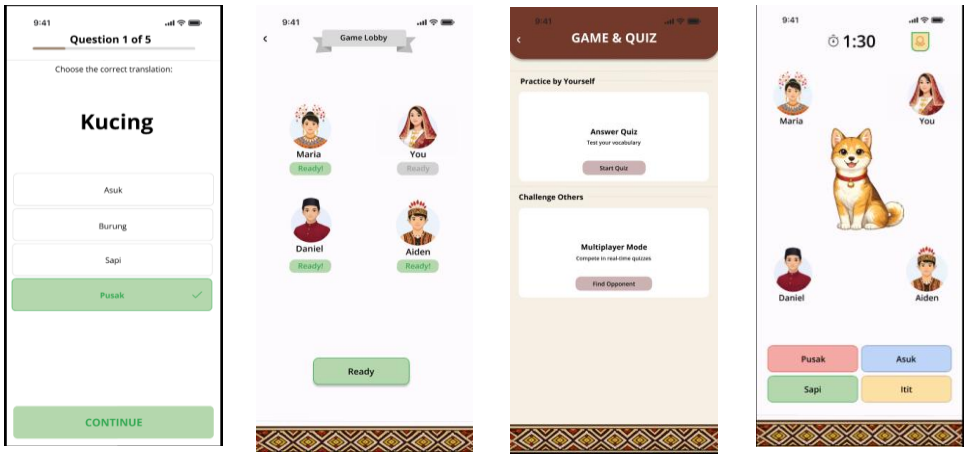

Figure 4: Game & Quiz Page

The Game & Quiz Page provides interactive learning activities where users can test their knowledge of Sarawak Malay vocabulary. These quizzes help reinforce what users have learned from the lessons through engaging practice exercises.

Users can choose to practice individually or challenge other players through the multiplayer mode. In the quiz interface, users are given questions related to Sarawak Malay vocabulary and must select the correct answer. The multiplayer feature allows users to compete with other players in real time, making the learning process more interactive and enjoyable.

Guest users must sign up before accessing this feature. Basic Plan users can participate in quizzes and games to practice their vocabulary, while Premium Plan users can also earn additional rewards such as bonus game coins and extra daily challenge coins through their gameplay.

4.5.5 Coin & Voucher System

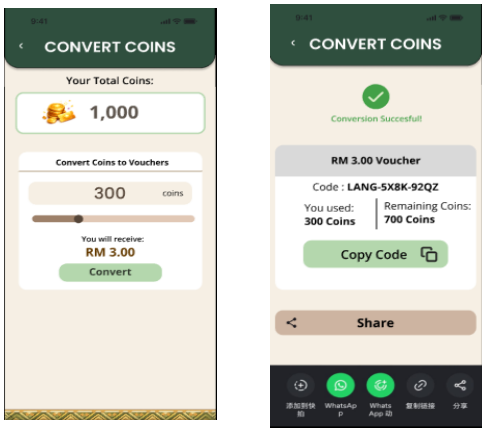

Figure 5: Coin & Voucher System

The Coin & Voucher System provides a reward mechanism that motivates users to actively participate in learning activities within the application. Users can earn coins by completing quizzes, games, and other learning tasks available on the platform.

The system allows users to convert their accumulated coins into voucher rewards. As shown in Figure 4.5, users can view their total number of coins and select the amount they wish to convert into vouchers. Once the conversion is completed, the system generates a voucher code that can be copied or shared with others

This reward mechanism encourages continuous participation in learning activities while increasing user engagement. By linking learning progress with tangible rewards, the application creates a more motivating and interactive learning experience for users.

5.0 Prototype Showcase

This section presents the prototype of SENADA, an AI-powered language learning platform designed to support the learning and preservation of Sarawak Malay. The prototype was developed using Figma to demonstrate the main features and user interface of the application. The following screenshots illustrate key components of the application and how users interact with the platform.

5.1 Get Started and Login

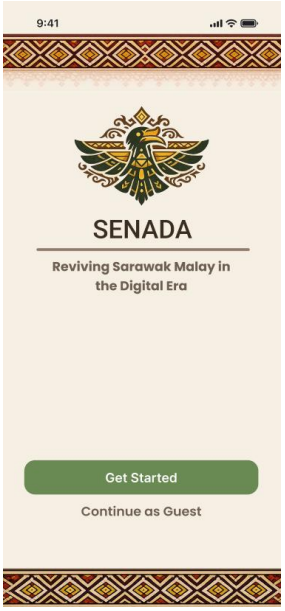

Figure 6: Get Started with Continue as Guest Interface

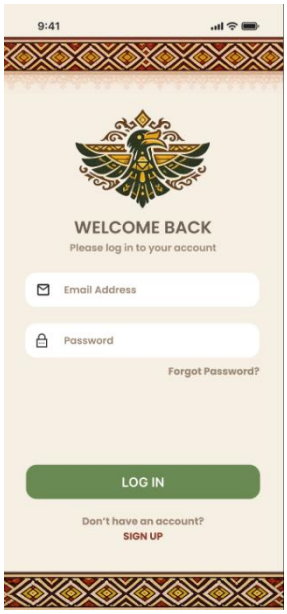

Figure 7: Login Interface

The Get Started and Login interface allows users to access the SENADA application. On the Get Started page, users are given the option to begin the login process or continue exploring the application as a guest. The "Continue as Guest" option is only available on the Get Started page and allows users to access limited features of the application without creating an account.

Users who choose to log in can enter their name and password to access their account. New users can create an account through the sign-up option by providing the required information. Existing users who have forgotten their password can recover their account using the "Forgot Password" feature, which guides them through the password recovery process.

5.2 Homepage Interface

Figure 8: Guest Homepage

Figure 9: Basic Plan Homepage

Figure 10: Premium Plan

The homepage serves as the main entry point of the SENADA application, providing users with quick access to the platform's key features and navigation options. From the homepage, users can access major sections of the application such as translation, learning modules, explore stories, communities, and other interactive features.

The content displayed on the homepage varies depending on the user's access level: Guest, Basic, or Premium. Guest users are presented with a simplified version of the homepage where several features are restricted. They are able to access limited translation, view the community section in read-only mode, and explore brief previews of cultural stories.

For Basic plan users, the homepage provides access to additional features such as games and quizzes, learning modules, full stories, and the ability to save selected words or phrases to favourites. The homepage layout remains similar but offers more interactive options compared to the Guest version.

Premium users have full access to all features directly from the homepage. This includes advanced translation options, podcast stories, unlimited favourites, and additional learning resources. The homepage therefore acts as a central hub that allows users to navigate the full range of features available in the SENADA platform.

5.3 Translation Interface

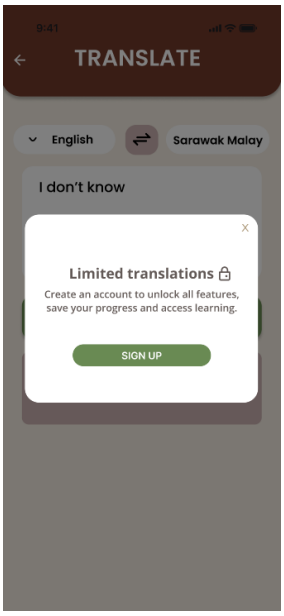

Figure 11: Limited Translation Interface for Guest

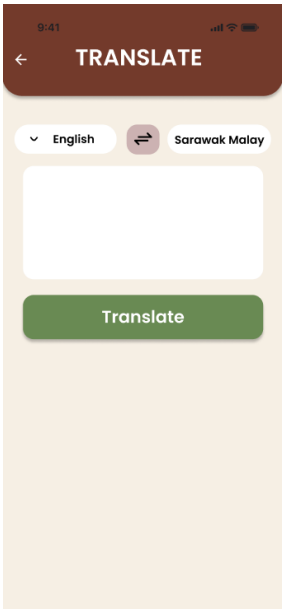

Figure 12: Translation Interface for Guest & Basic Plan

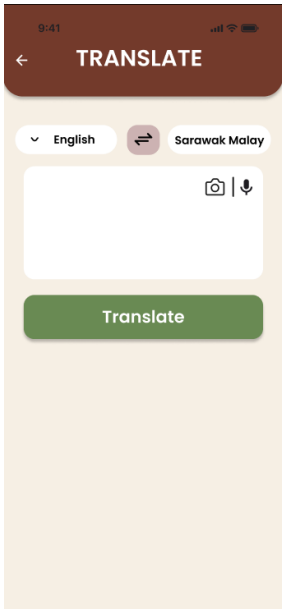

Figure 13: Translation Interface for Premium Plan (Camera & Voice)

Figure 14: Translation Interface for Premium Plan (With Ai Translation)

The translation interface allows users to translate Sarawak Malay words and phrases within the SENADA application. This feature supports language learning by helping users understand unfamiliar vocabulary quickly and conveniently.

Guest users are able to use the translation feature through text input only. However, the number of translations available to guest users is limited. This allows users to briefly explore the functionality of the application before signing up for a full account.

Basic plan users are able to perform unlimited text translations. The translations provided are simple translations that focus on the basic meaning of the word or phrase without detailed explanations.

Premium users have access to advanced AI-powered translation features. In addition to text translation, Premium users can translate words using voice input or camera scanning. The system also provides more detailed explanations of words and phrases, helping users better understand their meanings and usage in context. These additional features provide a more interactive and comprehensive language learning experience.

The translation interface therefore demonstrates the integration of AI technology in the SENADA platform while also reflecting the freemium access model implemented in the application

5.4 Learning Module Interface

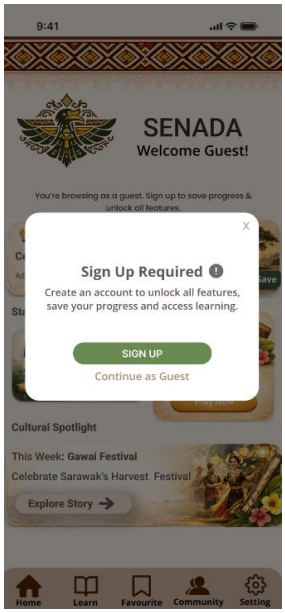

Figure 15: Sign Up Required for Guest to Access Learning Module

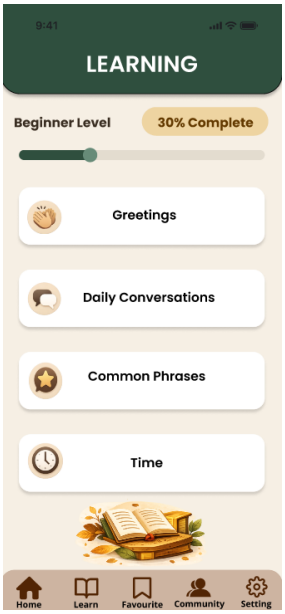

Figure 16: Learning Module Interface for Basic and Premium Plan

The Learning Module interface provides structured lessons that help users learn Sarawak Malay vocabulary and commonly used expressions. This section organizes learning content into several categories, allowing users to navigate different topics in a clear and systematic manner.

The learning modules include categories such as greetings, daily conversations, common phrases, and time expressions. Each category introduces users to commonly used words and phrases together with their meanings and usage in everyday communication. This structured approach helps users gradually develop their understanding of the language.

Access to the Learning Module is available only to Basic and Premium users. Guest users cannot access this section of the application. This design encourages users to create an account to access the learning materials and fully engage with the language learning features provided by the SENADA platform.

5.5 Game & Quiz Interface

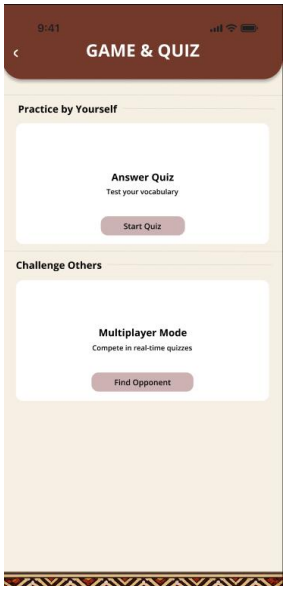

Figure 17: Game & Quiz Interface for Basic & Premium Plan

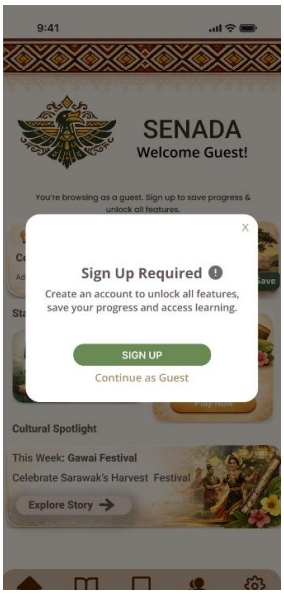

Figure 18: Sign Up Required for Guest to Access Game & Quiz

The Game and Quiz interface provides users with an interactive way to practice and reinforce their understanding of Sarawak Malay vocabulary and expressions. This feature allows users to test their knowledge through quiz activities and language learning challenges.

Guest users cannot access the Game and Quiz feature and are required to sign in or create an account to participate in these activities.

Basic and Premium users can access the Game and Quiz feature. The interface includes both a practice mode, where users can complete quizzes individually, and a multiplayer mode, where users can challenge other players. This interactive approach encourages users to actively practice the language while engaging with other learners.

Users can earn coins by completing quizzes and challenges. The coins collected can later be converted into vouchers within the platform. These vouchers may include rewards such as discounts or promotional vouchers from partnered services, for example food delivery or ride-hailing platforms. This reward system provides additional motivation for users to continuously engage with the learning activities.

5.6 Explore Stories Interface

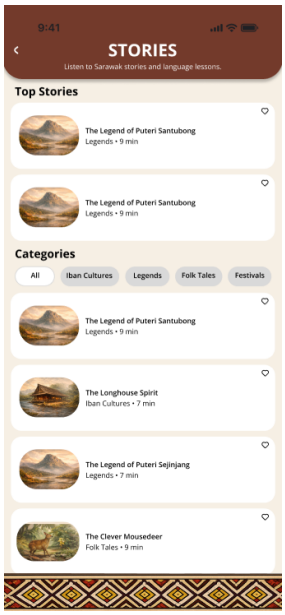

Figure 19: Explore Stories

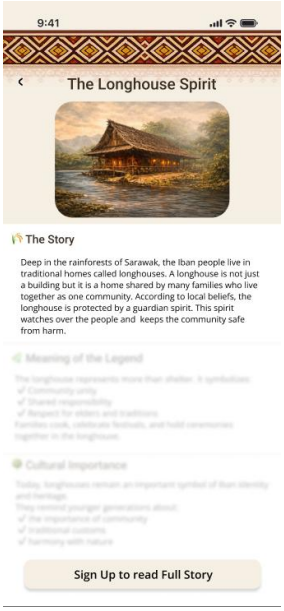

Figure 20: Story Preview for Guest

Figure 21: Full Story for Basic Plan

Figure 22: Full Story with Podcast for Premium Plan

The Explore Stories interface introduces users to cultural stories related to Sarawak and its local traditions. This section allows users to learn the Sarawak Malay language while also understanding the cultural context behind certain expressions and narratives.

Guest users are able to access the Explore Stories section; however, they can only view a brief preview of each story. To read the full story, guest users are required to sign in or create an account.

Basic plan users are able to read the full stories available within the platform. This allows them to explore cultural content while improving their understanding of Sarawak Malay vocabulary and expressions.

Premium users have full access to the stories as well as additional features such as podcast versions of the stories. The podcast feature allows users to listen to the stories in audio form, providing a more engaging learning experience and helping users improve their pronunciation and listening skills.

The Explore Stories feature therefore combines language learning with cultural storytelling, supporting the preservation and appreciation of Sarawak's linguistic and cultural heritage.

5.7 Community Interface

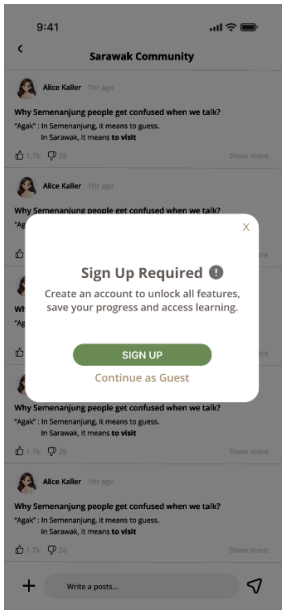

Figure 23: Sign Up Required to Interact with Communities for Guest

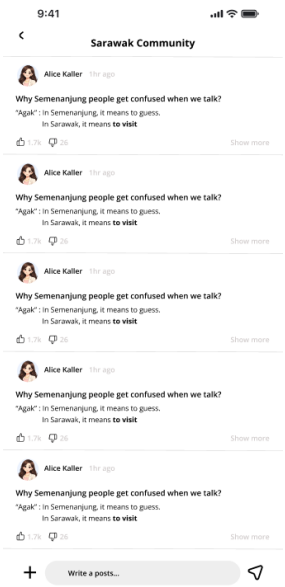

Figure 24: Write a Post for Basic and Premium Plan

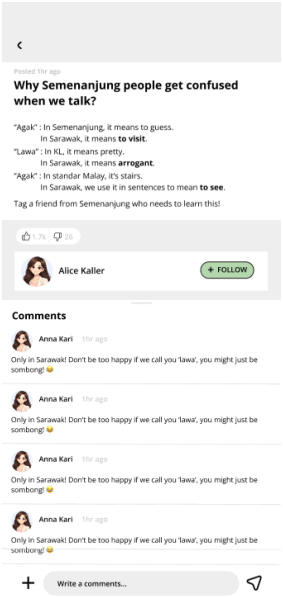

Figure 25: Write a Comment, Like & Follow for Basic and Premium Plan

The Community interface allows users to interact with other members of the SENADA platform and share their language learning experiences. This section creates a collaborative environment where users can engage in discussions related to Sarawak Malay vocabulary, cultural expressions, and learning progress.

Guest users are able to access the community section in read-only mode. They can view posts and discussions within the community but are not able to interact with other users.

Basic and Premium users have full access to the community features. They are able to create posts, like and comment on other users' posts, as well as follow other members within the platform. These interactive features encourage user engagement and allow learners to connect with others who are also interested in learning Sarawak Malay.

The community feature helps build a supportive learning environment where users can exchange knowledge, share experiences, and practice the language together.

5.8 Premium Plan / Subscription Interface

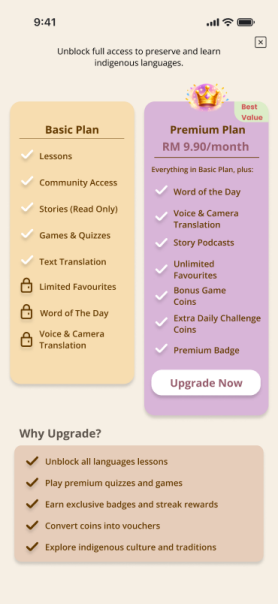

Figure 26: Subscription Interface

The Premium Plan interface allows users to upgrade their account to access additional features within the SENADA platform. This section presents the available subscription plan and highlights the benefits offered to Premium users.

The SENADA platform adopts a freemium model, where users can access basic features for free while advanced functionalities are available through a Premium subscription. The Premium plan is priced at RM9.90 per month and unlocks several enhanced features designed to improve the overall learning experience.

Premium users gain access to features such as Word of the Day with pronunciation guidance and the option to add the word directly to favourites. They can also use advanced translation tools including text, voice, and camera translation, along with more detailed AI-generated explanations. In addition, Premium users are able to listen to podcast versions of stories in the Explore Stories section, providing a more interactive way to learn the language.

Premium users also receive additional rewards through the platform's gamification system. They can earn bonus coins from the Game and Quiz feature and receive extra coins when completing daily challenges. Furthermore, Premium users can save an unlimited number of words or phrases to their favourites and receive a Premium badge that distinguishes their account within the platform.

This interface clearly presents the advantages of upgrading to the Premium plan while encouraging users to unlock the full capabilities of the SENADA application.

6.0 AI Acknowledgement

Artificial Intelligence (AI) tools were used during the development of this project to support research, idea generation, and writing assistance. This section provides transparent disclosure of the AI tools used in the development process. These tools helped the team explore possible solutions, refine explanations, and organize the report more effectively. However, AI tools were only used as supporting tools, and all outputs were carefully reviewed, edited, and verified by the team members to ensure accuracy and originality.

The following AI tools were used during the project development process:

6.1 ChatGPT (OpenAI)

ChatGPT was used to assist with brainstorming project ideas, improving sentence structure, and refining explanations in several sections of the report. It also helped the team explore possible features for the proposed AI-powered language learning platform and provided guidance on organizing report content more clearly. In addition, ChatGPT was used to generate certain visual assets for the prototype, such as icons and badges used in the application interface. All responses and generated materials provided by ChatGPT were reviewed and adjusted by the team to ensure that the final content reflects the team's own understanding and interpretation.

6.2 Gemini (Google)

Google Gemini was used as an additional AI support tool during the development of the project. It was mainly used to assist with researching background information, generating alternative explanations, and clarifying concepts related to artificial intelligence and language learning applications. Similar to ChatGPT, the suggestions provided by Gemini were carefully evaluated and edited by the team members before being incorporated into the final report.

6.3 Responsible Use of AI

The team ensured that AI tools were used responsibly and ethically throughout the project development process. AI tools were used only as supportive resources and did not replace the team's original thinking, analysis, or decision-making. All information generated by AI tools was carefully reviewed and verified by the team members to ensure reliability and accuracy.

In addition, the team ensured that AI-generated outputs were evaluated to avoid misleading information or potential bias. No sensitive or personal data was shared with AI tools during the development process. The final report, prototype design, and project ideas therefore represent the collective work and understanding of the team members.

7.0 Conclusion

This project aims to address the limited digital representation of Sarawak Malay by proposing an AI-powered language learning and translation platform. Although Sarawak Malay is widely used in everyday communication in Sarawak, it remains underrepresented in modern digital tools such as translation systems and language learning applications. This lack of digital support restricts opportunities for both local communities and outsiders to learn and understand the language, which may gradually reduce its presence in digital spaces.

By integrating artificial intelligence with interactive language learning features, the proposed platform aims to make Sarawak Malay more accessible while also supporting the preservation of local linguistic and cultural heritage. The application not only provides translation functionality but also introduces users to cultural meanings, expressions, and pronunciation of the dialect. Through this approach, the project contributes to the promotion of inclusive education and cultural awareness while supporting the United Nations Sustainable Development Goals (SDG 4: Quality Education, SDG 10: Reduced Inequalities, and SDG 11: Sustainable Cities and Communities).

7.1 Stakeholders

Several key stakeholders are important in ensuring the successful implementation and long-term impact of this project.

One of the main stakeholders is the **Sarawak government and tourism authorities**, such as the Sarawak Tourism Board. The government can help promote the application to visitors entering Sarawak through promotional materials such as flyers, banners, and QR codes displayed at locations like Kuching International Airport and tourist information centers. Through this collaboration, international and domestic visitors can easily discover and download the application. Tourists can use the app not only for translation purposes but also to learn basic Sarawak Malay phrases and understand the cultural meanings behind certain expressions, which can improve communication and cultural appreciation during their visit.

Another important stakeholder group is **professional tour guides and tourism companies**, such as Amazing Borneo Tours and GoWithGuide. Tour guides can incorporate the application into their tours by introducing visitors to common Sarawak Malay phrases and cultural expressions during guided activities. The application can also include tourism-related language modules that feature commonly used phrases in markets, restaurants, and cultural sites. This

initiative can enhance the digital tourism experience in Sarawak by helping visitors engage more deeply with the local language and culture. Through this collaboration, the application can function not only as a translation tool but also as a cultural learning platform that enriches tourists' overall travel experiences.

Educational institutions such as schools and universities also play an important role in supporting the sustainability of the project. Both schools and universities can use the application as a supplementary learning tool to introduce students to Sarawak Malay vocabulary, pronunciation, and cultural stories. The platform can support language learning through interactive features that help students understand the dialect in a more engaging way. In addition, universities can contribute to the development of the platform through research and language documentation. Institutions such as Universiti Malaysia Sarawak (UNIMAS), Swinburne University Sarawak, and Curtin University Malaysia may assist in linguistic research, data collection, and the validation of Sarawak Malay vocabulary and expressions. This collaboration can help ensure that the language content within the platform remains accurate, educational, and culturally appropriate while also supporting academic research on indigenous languages.

Local communities and native speakers of Sarawak Malay are also essential stakeholders, as they provide authentic language knowledge, pronunciation, and cultural context that are necessary for building a reliable and culturally meaningful language database.

7.2 Scalability Plan

Although the current project focuses on Sarawak Malay, the platform is designed with scalability in mind so that it can support additional languages in the future. The system can gradually expand to include other indigenous languages and dialects in Sarawak, such as Iban, Bidayuh, and Melanau. By expanding the language database and AI training datasets, the platform can become a broader language learning ecosystem that supports multiple local languages.

In the long term, the platform has the potential to expand beyond Malaysia and support indigenous languages across Southeast Asia. Many ASEAN countries face similar challenges in preserving regional languages and dialects that lack digital representation. By adopting a modular and expandable language framework, the application could eventually support cross-cultural learning and digital language preservation across the ASEAN region.

7.3 Sustainability Plan

To ensure long-term sustainability, the project can continue developing through collaborations with government agencies, educational institutions, tourism organizations, and local communities. Continuous contributions from native speakers can help expand and enrich the language database by adding new vocabulary, phrases, and cultural stories. This community-driven approach can help preserve oral traditions and ensure that the language remains relevant for future generations.

In addition, partnerships with tourism organizations and educational platforms can help maintain the application's usage and visibility. The integration of AI technologies will allow the platform to improve over time as more language data becomes available. With consistent updates, research collaboration, and community participation, the application can evolve into a sustainable digital platform that supports the preservation of indigenous languages while promoting cultural understanding and inclusive technological development.

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